Formative Task 'Discussion Forum: AI, Blockchain, and Quantum Computing'

Unit 12

The Future of AI, Blockchain and Quantum Computing in Cloud Operations

Cloud computing is developing beyond simply applications delivery and data storage. it is likely that in the future, and even now, cloud operations are and will become more automated, secure and capable of handling more specialized computing tasks. Three technologies that could serve as the foundation for such a development are; artificial intelligence (AI), blockchain and quantum computing.

AI plays a pivotal role because cloud environments frequently adapt according to user demand. Machine learning can help predict workloads usage and determine when computing resources need to be adapted efficiently. Some research indicates that workload forecasting can help to improve resource distribution and service reliability in cloud environments.

(Rossi et al., 2025).

Open-source projects like the TensorFlow are important because they provide developers with the tools to build and deploy machine-learning applications without building an AI framework entire from the beginning.

Using Blockchain few different problems could be addressed: trust. Cloud services usually involve data sharing and computing resources between multiple organizations; Blockchain could provide shared and traceable records of transactions and activities. some research has explored blockchain for managing virtual machines across multi-

cloud environments, demonstrating strong potential for improved communication and resource management.

(Altahat, Daradkeh and Agarwal, 2025).

For instance, Hyperledger which uses its open-source blockchain technologies can be adapted by organizational and enterprise applications.

Quantum computing is still developing, though its relationship to cloud computing is already becoming important. Cloud access lets organizations test and experiment with quantum computers without the need to own costly and specialized hardware. Qiskit helps such development by providing open-source tools for the creation and use of quantum applications.

(Javadi-Abhari et al., 2024).

The most interesting future development could be the combination of these technologies. AI could manage cloud resources, blockchain could provide trust and transparency, and quantum computing could provide specialized technology to solve problems that are too difficult for conventional computers. Open-source projects such as TensorFlow, Hyperledger and Qiskit can accelerate this development by providing researchers and developers tools to experiment, collaborate and build on existing work.

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